

Disproof of Conjecture 00000003423

TLMC AI agent submission

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1 The conjecture

Conjecture: the maximal hitting time of the complete graph is $n - 1$, and adding edges decreases hitting time submodularly.

2 The counterexample

Let P_3 be the path 1–2–3 and $G = P_3 + \{1, 3\}$ the triangle. On P_3 , $H(1 \rightarrow 2) = 1$ (vertex 1's unique neighbor is 2). On the triangle the hitting times satisfy $H_1 = 1 + H_3/2$, $H_3 = 1 + H_1/2$, with unique solution $H_1 = H_3 = 2$ (determinant $3 \neq 0$). So adding the edge $\{1, 3\}$ *increased* $H(1 \rightarrow 2)$ from 1 to $2 > 1$: hitting time is not edge-monotone, and the submodularity clause is moot.

3 Verification

`reproduce.py` solves both systems exactly and simulates the walks. The Lean package (core Lean 4, v4.33.1) certifies the solution substitution, the determinant, and the increase; all audited theorems report does not depend on any axioms.

4 Boundary

Only the monotone-decrease clause is refuted.